

Glyph maps for spatio-temporal data in R

Seeing seasonality at every location with **sugarglider**, **cubble** and **GGally**

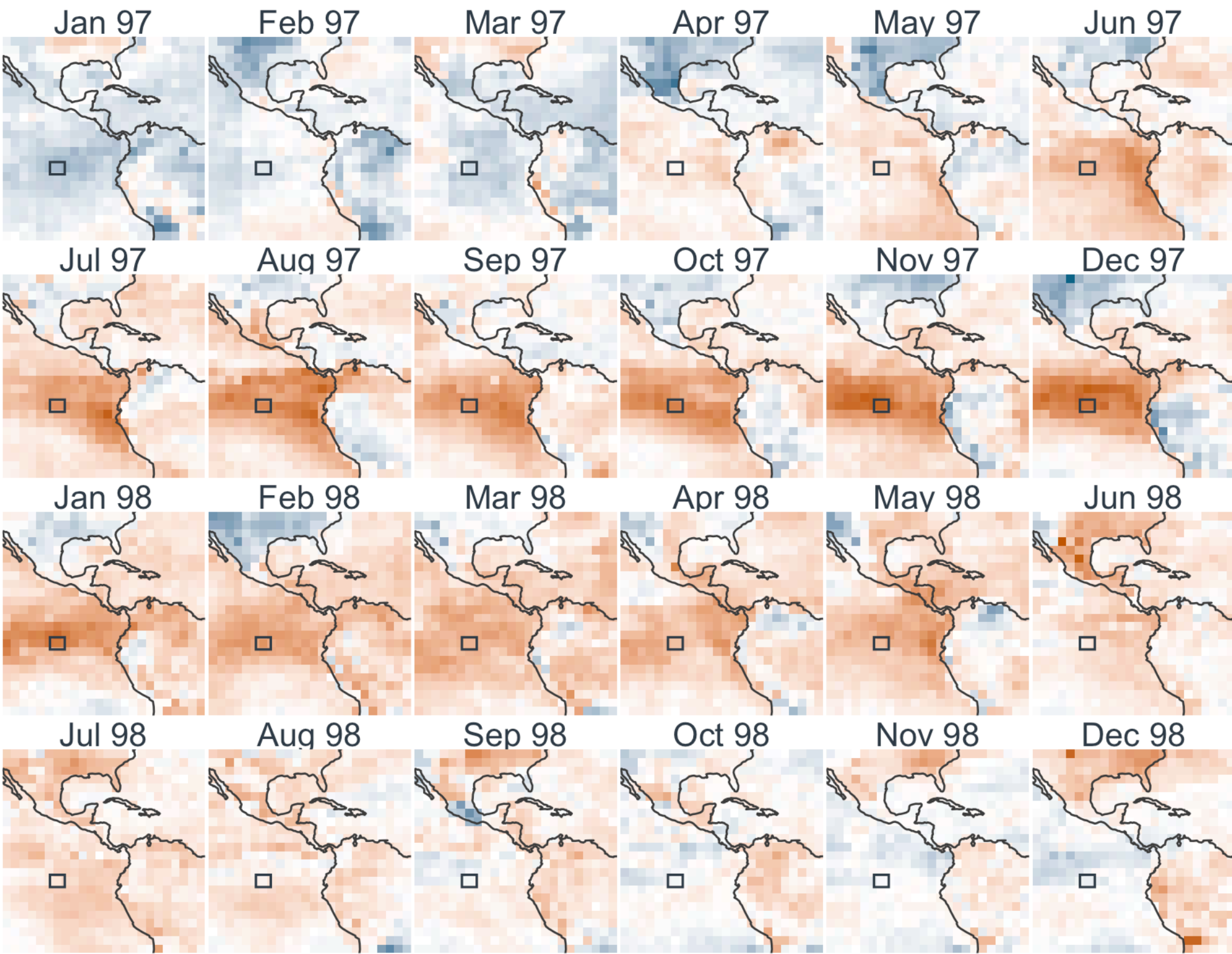
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2012 From facets to a glyph map

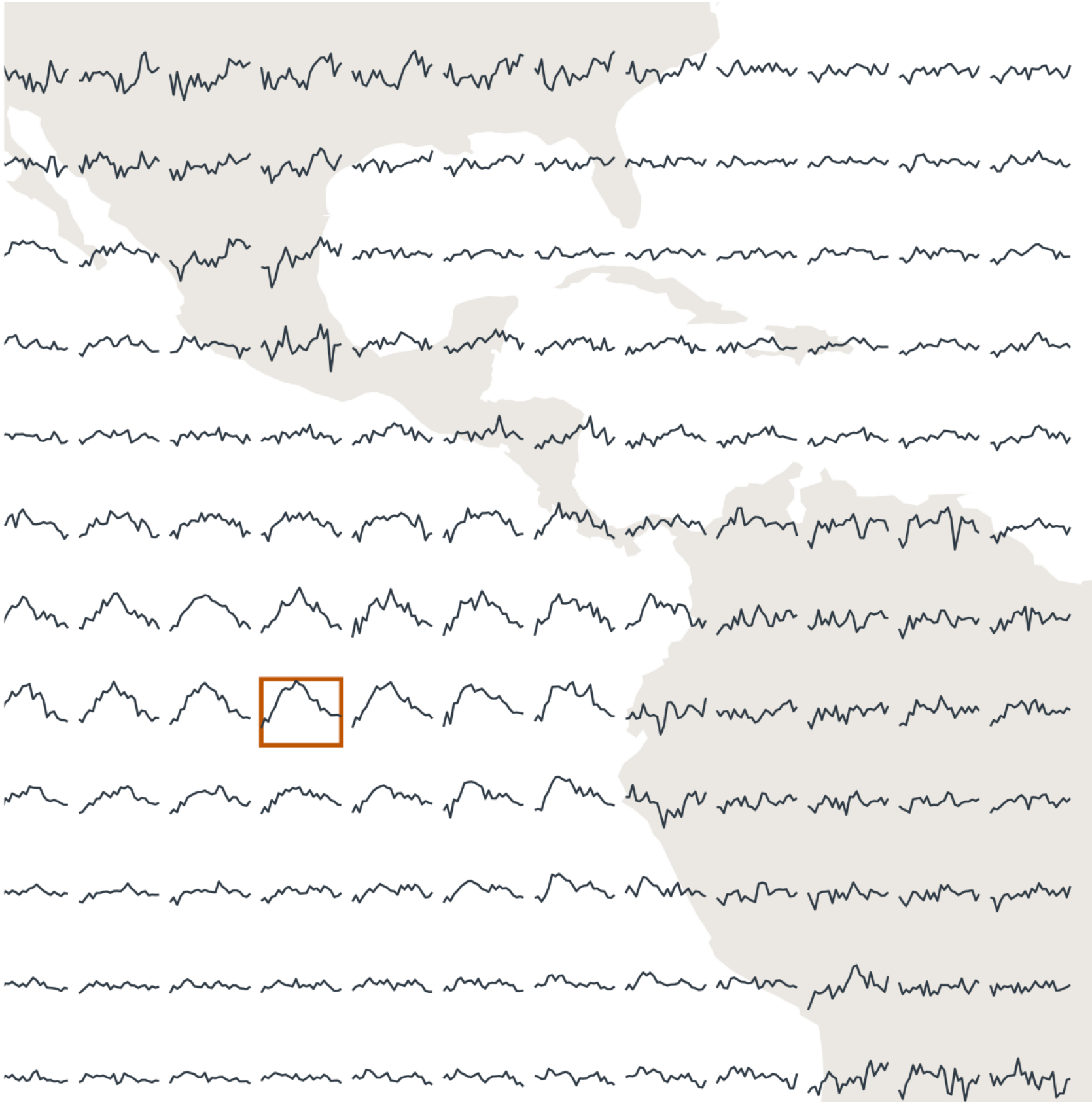
Environmental data often come as **many time series at many locations**, commonly shown as maps faceted by time. Below, monthly de-seasonalised surface temperature over Central America for 1997–98.



Trace one location. Follow the boxed cell through the 24 panels to get its time series. A **glyph map** draws that series as a small plot at the cell's location (Wickham et al., 2012).



Do this for every location to get the full glyph map:

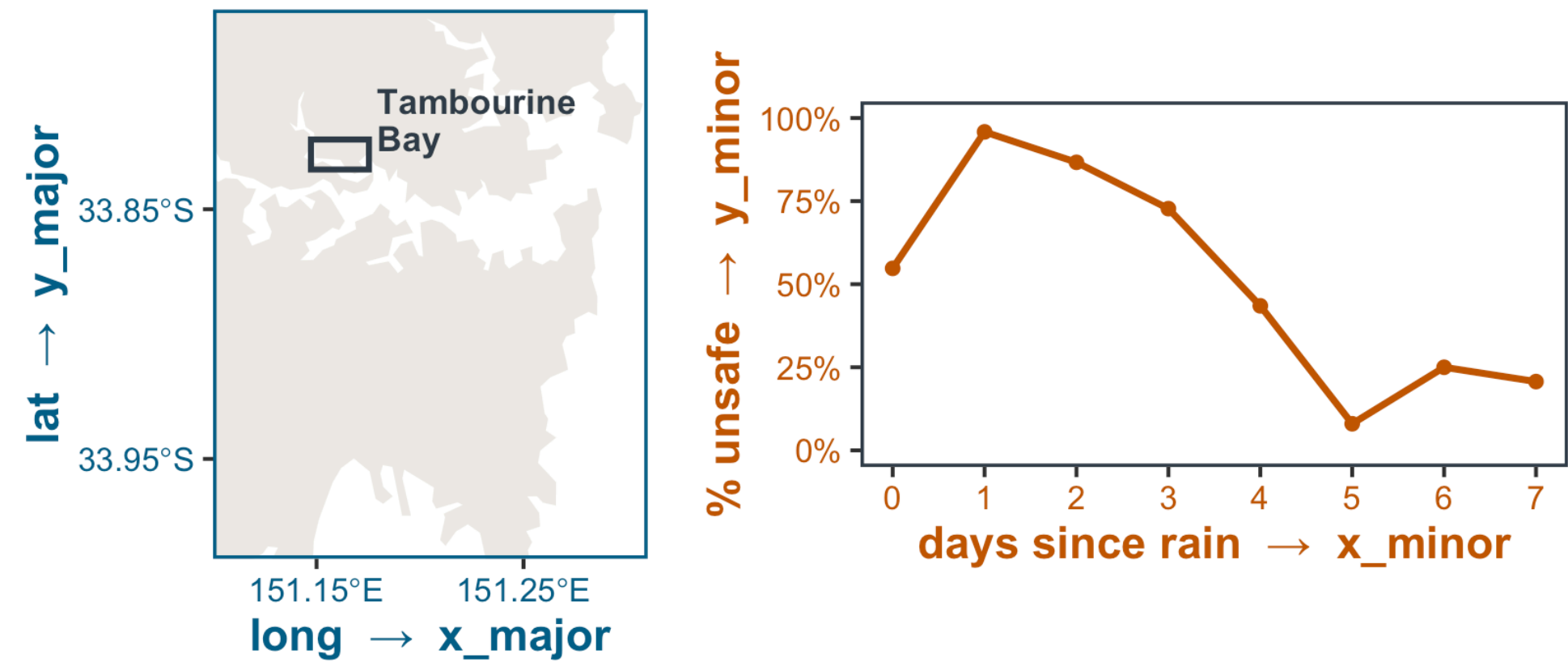


What the glyph map shows:

- Equatorial eastern Pacific glyphs share one hump peaking in late 1997 (the El Niño), fading away from the equator.
- Glyphs over land vary far more month to month than those over the sea.

2023 Four new ggplot2 aesthetics

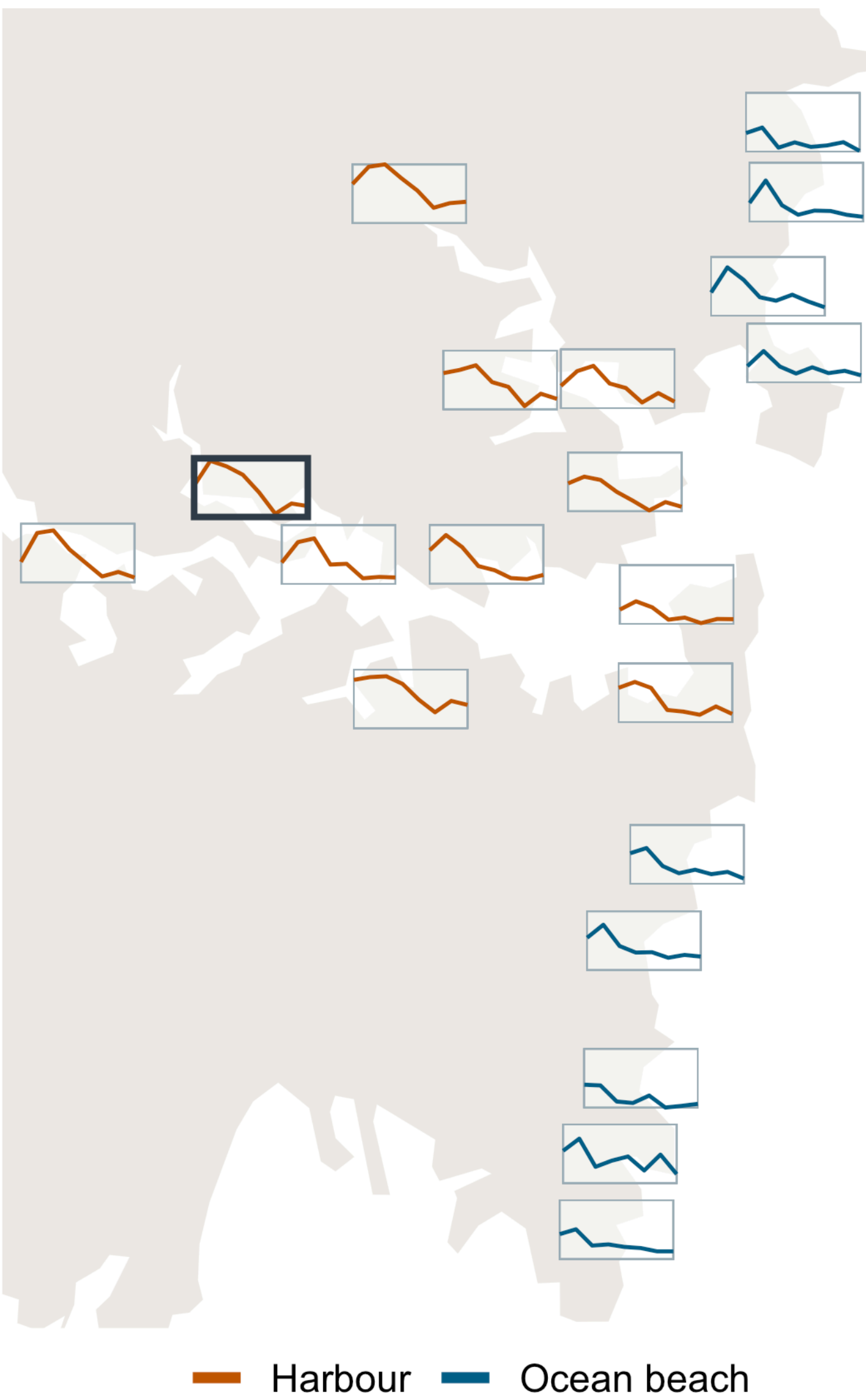
cubble implements the glyph-map transformation in ggplot2 syntax: `geom_glyph()` takes the major and minor coordinates as four new **aesthetics**, so making a glyph map is a matter of mapping your variables onto them. Here: **how long after a storm is it safe to swim in Sydney?** Share of water samples above the bacteria guideline (40 enterococci/100 mL) in the 7 days after ≥ 20 mm of rain, 1993–2025 (TidyTuesday 2025-05-20).



Tambourine Bay's location gives the **major** coordinates; its recovery curve after rain gives the **minor** coordinates drawn inside its box.

Because the glyph is an ordinary ggplot2 layer, it pieces together with other layers. The key parts of the code (simplified):

```
ggplot(beach,
  aes(x_major = long, y_major = lat,
      x_minor = days, y_minor = unsafe,
      colour = water)) +
  geom_sf(data = sydney, inherit.aes = FALSE) + # map
  geom_glyph_box(width = 0.028, height = 0.012) +
  geom_glyph(width = 0.028, height = 0.012) +
  scale_colour_manual(values = ...)
```



- Ocean beaches** spike the day after a storm (46% of samples unsafe), then clear within 2 days (24%).
- Harbour** sites are worse and slower: 73% unsafe the day after, still 65% two days later, especially in the inner-harbour coves.

2026 Multivariate glyphs

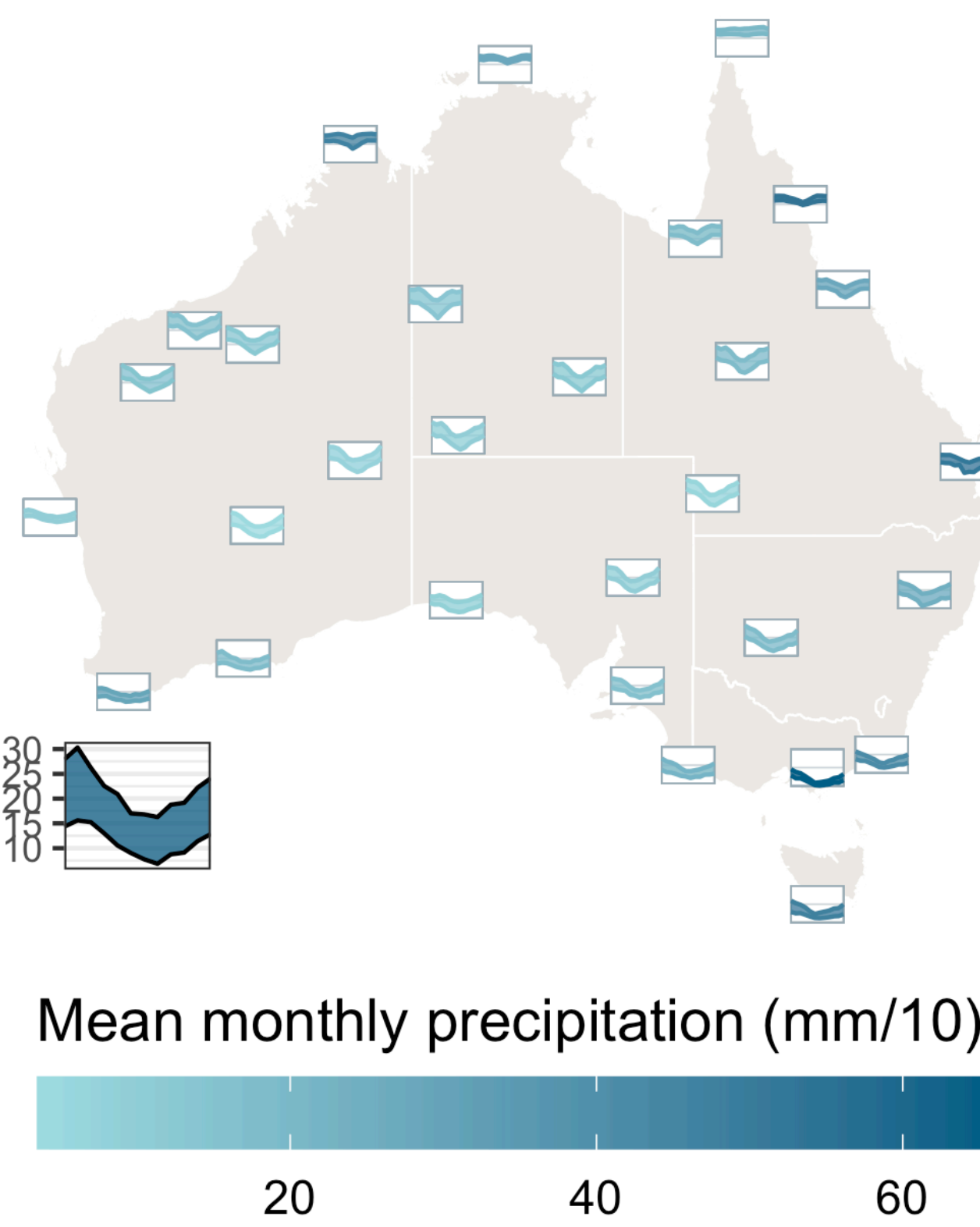
A line glyph shows **one variable** over time, but many environmental summaries are **multivariate**: daily minimum *and* maximum, lower *and* upper quantiles, an estimate *and* its uncertainty. **sugarglider** adds aesthetics for a second minor variable, transformed the same way:

$$y_{\min} = y_{\text{major}} + \frac{h}{2} \tilde{y}_{\min, \text{minor}}$$

$$y_{\max} = y_{\text{major}} + \frac{h}{2} \tilde{y}_{\max, \text{minor}}$$

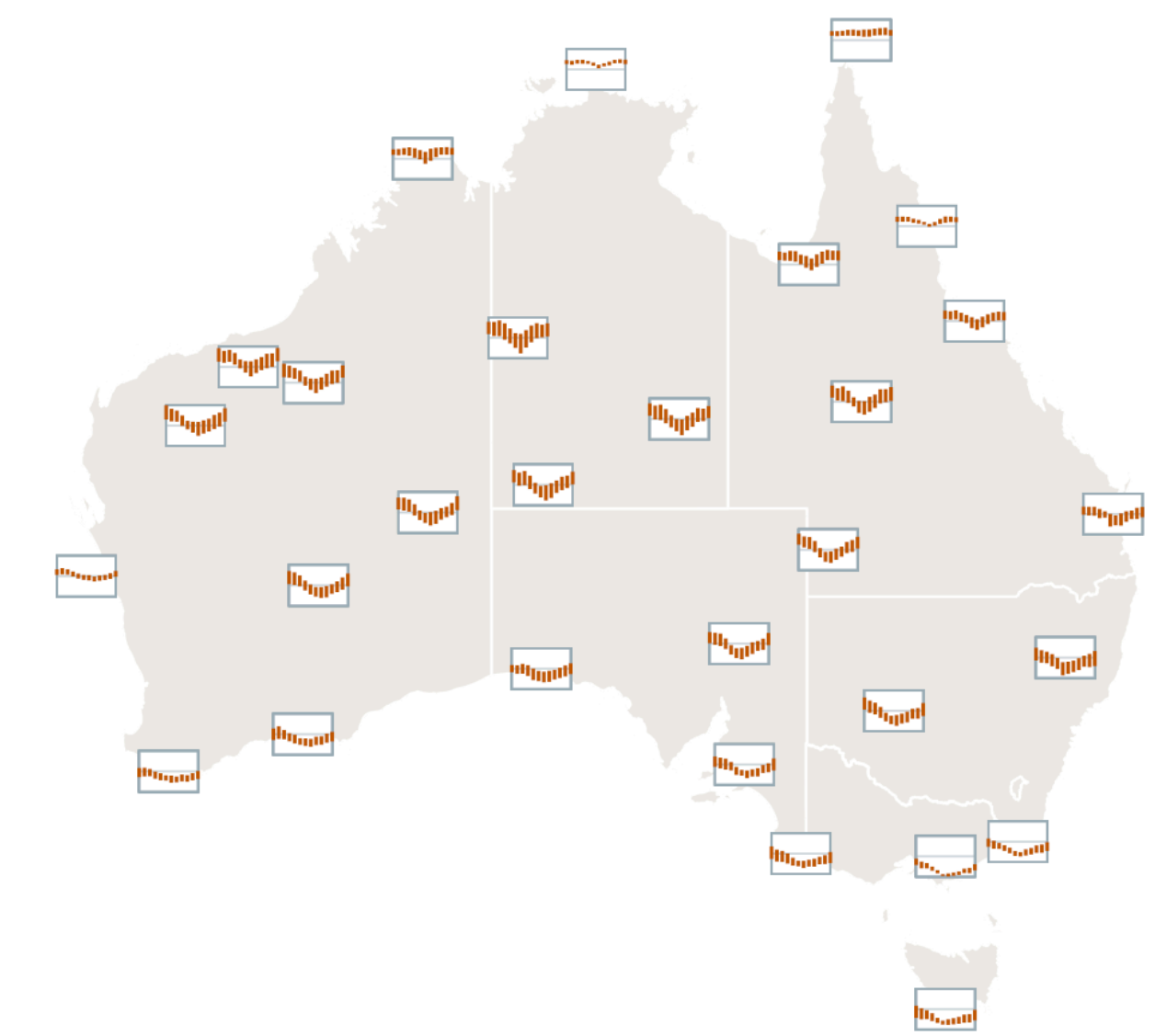
Ribbons fill the band between the two, here with a third variable, rainfall, as colour and an enlarged legend glyph to read the scale:

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat,
             x_minor = month,
             ymin_minor = tmin,
             ymax_minor = tmax)) +
  add_glyph_boxes() + add_ref_lines() +
  add_glyph_legend() +
  geom_glyph_ribbon(aes(fill = prcp))
```



Segments draw one bar per time point from the same data:

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat,
             x_minor = month,
             y_minor = tmin, yend_minor = tmax)) +
  add_glyph_boxes() + add_ref_lines() +
  geom_glyph_segment()
```



Wet tropical stations have narrow, flat bands; inland stations show wide daily ranges and deep winter dips.

Which one should I use?

	GGally	cubble	sugarglider
Interface	<code>glyphs()</code> → data frame	ggplot2 layer	ggplot2 layer
Glyph geometry	line	line	ribbon, segment
Reference box & line	✓	✓	✓
Global / local rescale	via <code>y_scale</code>	✓	✓
Polar glyphs	✓	✓	–
Glyph legend & theme	–	–	✓
Spatio-temporal data structure	–	✓	via cubble
Interactive (ggraph, leaflet)	–	via crosstalk	✓

Take-home

- A glyph map shows each site's **temporal shape** and the **spatial pattern** in one plot.
- If your summary is a **range** (min–max, quantiles, uncertainty), use sugarglider's **ribbon** or **segment** glyphs.
- All three packages work with **ggplot2**, so you can add base maps, colour and facets as usual.

Get started

```
install.packages(
  c("sugarglider", "cubble", "GGally"))
```

Paper



sugarglider



cubble



GGally



Wickham, Hofmann, Wickham & Cook (2012). Glyph-maps for visually exploring temporal patterns in climate data and models. *Environmetrics* 23(5), 382–393. doi:10.1002/env.2152
Zhang, Cook, Laa, Langrené & Menéndez (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *J. Stat. Softw.* 110(7). doi:10.18637/jss.v110.i07
Po, Yang, Zhang & Cook (2026). sugarglider: Create glyph-maps of spatiotemporal data. *The R Journal* 18(3), 286–304. doi:10.32614/RJ-2026-050
Beach data: TidyTuesday 2025-05-20.